Machine Learning Project Report

Car Price Prediction and Football Player Scouting System

Prepared from the provided Jupyter notebook. The report summarizes the datasets, preprocessing, models, evaluation results, and final system behavior using details from the project itself.

Project Area	Main Objective	Dataset Size	Final Output
Assignment 1: Car Price	Predict car prices and classify price levels	72,435 rows, 10 original columns	Regression model + KNN classifier
Assignments 2-3: Football Players	Predict player market value and performance tier	19,667 rows, 9 columns	Final scouting system with regression and classification

1. Executive Summary

- The notebook contains a complete applied machine learning workflow: data loading, exploratory analysis, preprocessing, feature engineering, model training, model comparison, and final system testing.
- The first part focuses on used car price prediction. It handles missing values, creates new features such as car_age and km_per_year, applies one-hot encoding, and evaluates both regression and classification models.
- The second and third parts focus on a football player dataset. Assignment 2 builds baseline regression and classification models, while Assignment 3 improves the system using KNN, SVM, Random Forest, and ensemble methods.
- The final football scouting system selected a Voting Regressor Ensemble for market value prediction and an SVM Classifier for performance tier prediction.

2. Dataset Overview

2.1 Car Price Dataset

The car dataset starts with **72,435 rows and 10 columns**. The original features include model, year, price, transmission, mileage, fuelType, tax, mpg, engineSize, and Make. The target variable for regression is price. The project also creates price classes for classification.

Important Detail	**Value / Explanation**
Original shape	72,435 rows x 10 columns
Dropped column	model, because it had 145 unique values and was considered less useful for the model
Engineered features	car_age = 2026 - year; km_per_year = mileage / car_age
Missing values	Most columns had around 5% missing values; km_per_year had about 9.70% after feature engineering
Train/test split	55,051 training samples and 13,763 test samples after dropping rows with missing price

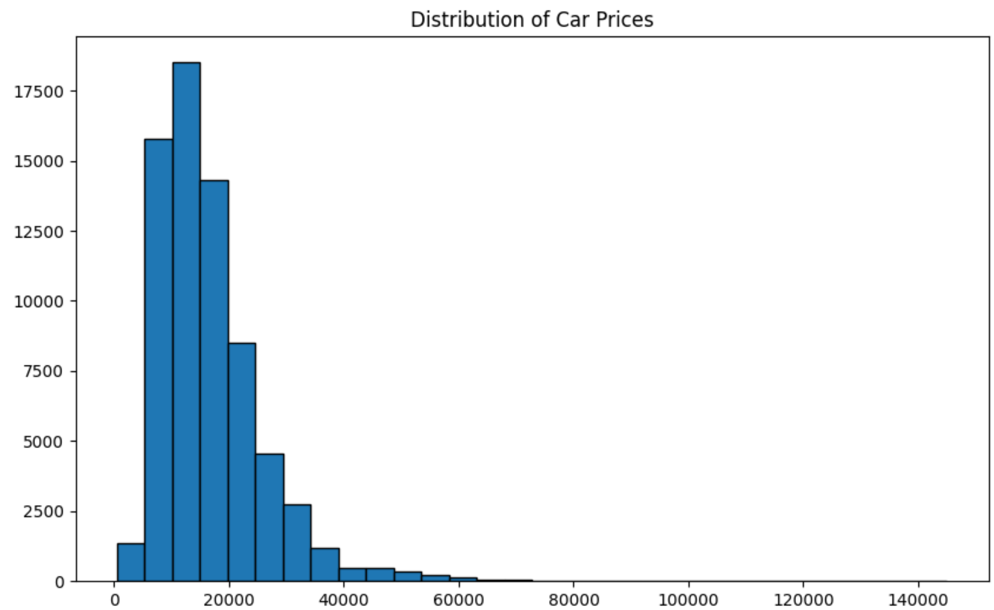

Figure 1. Distribution of car prices. The notebook reports a right-skewed distribution with skewness = 2.1809.

2.2 Football Player Dataset

The football dataset contains **19,667 rows and 9 columns**. The project uses Value Per M$ as the regression target and creates Performance_Tier from Overall_Rating quartiles for classification.

Column Group	Examples
Player identity / categorical	Name, Country, Position, Team
Numerical predictors	Age, Overall_Rating, Future Potential, Total_Stats Score
Regression target	Value Per M$
Classification target	Performance_Tier: Low, Mid, High, Elite
EDA Finding	**Project Result**
Missing values	0 missing values in all columns
Value distribution	Strongly right-skewed; skewness = 7.9832
Strongest correlation with market value	Overall_Rating = 0.5606
Next strongest correlations	Future Potential = 0.5010; Total_Stats Score = 0.3851; Age = 0.1423
Largest outlier percentage	Value Per M$ had 2,390 outliers, around 12.15% by IQR

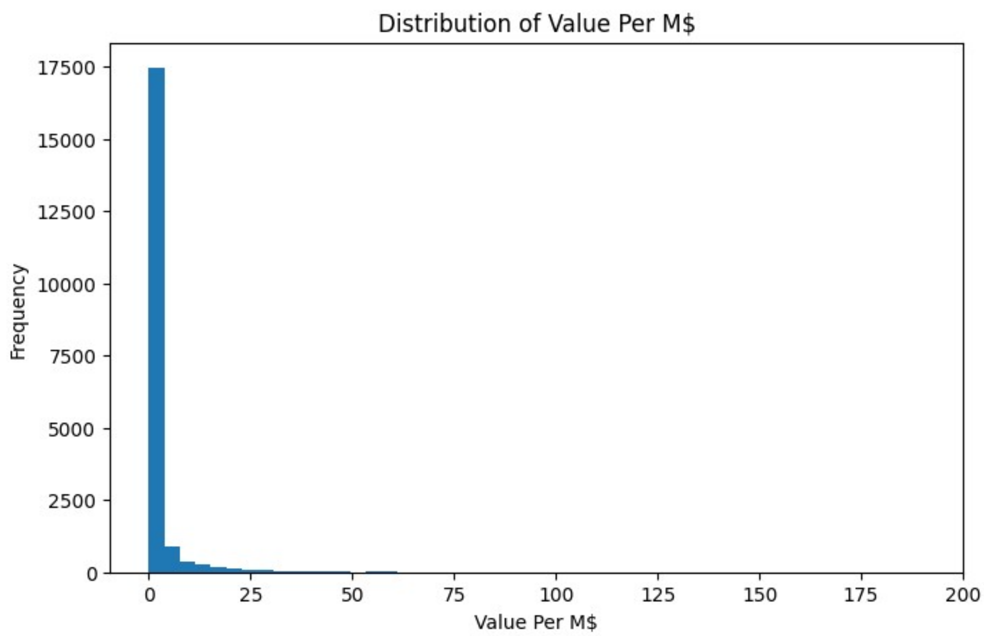

Figure 2. Distribution of player market value in millions of dollars.

3. Preprocessing and Feature Engineering

Step	Car Price Project	Football Project
Missing values	Dropped missing price rows; numerical median imputation; categorical mode imputation	No missing values found, so no imputation was needed for the raw dataset
Outliers	IQR-based outliers in car_age, km_per_year, tax, mpg, and engineSize were replaced with median; y_train was clipped at 1st and 99th percentiles	IQR outliers were reported; Value Per M$ was highly skewed
Encoding	OneHotEncoder for transmission, fuelType, and Make	OneHotEncoder for categorical features such as Country, Position, and Team
Scaling	StandardScaler was used for numerical features	Pipelines used scaling where needed for models such as Logistic Regression, SVM, and KNN
Feature engineering	car_age and km_per_year	Performance_Tier from Overall_Rating quartiles

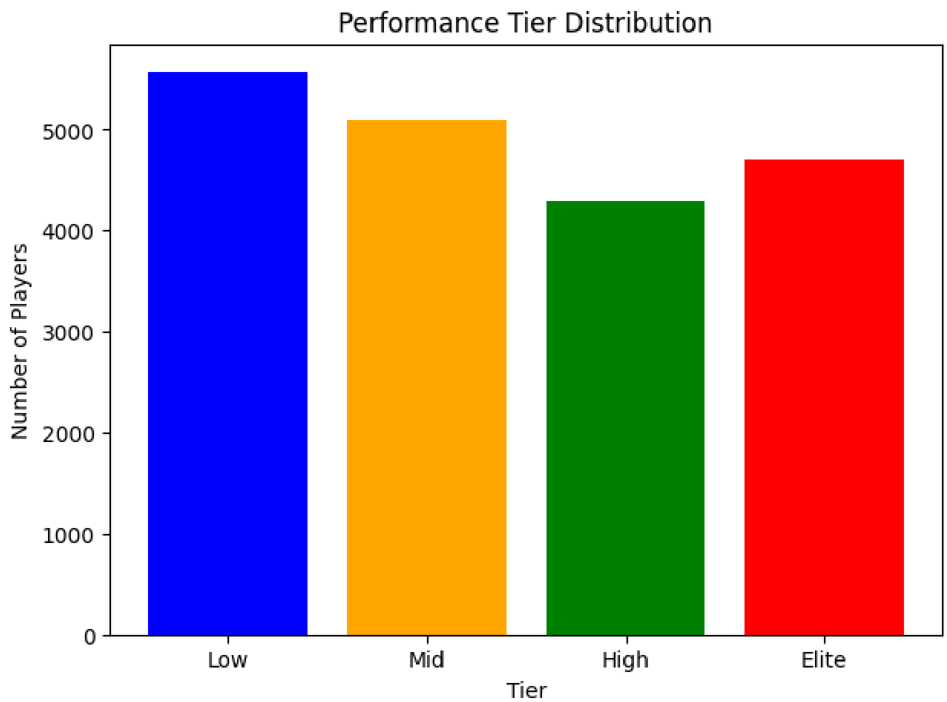

Figure 3. Performance tier distribution after creating the classification target.

4. Assignment 1: Car Price Modeling Results

Model / Task	Main Method	Result
Regression	Linear Regression with log-transformed target	R2 = 0.7971; RMSE = 4,155.27
Classification target	Cheap / Moderate / Expensive based on price quantiles	Used to convert the price problem into a class prediction task
KNN tuning	GridSearchCV over neighbors 1-30 and euclidean/manhattan distance	Best parameters: n_neighbors = 4, metric = 'manhattan'
KNN test performance	Weighted metrics	Accuracy = 0.6214; Precision = 0.6214; Recall = 0.6214; F1 = 0.6204

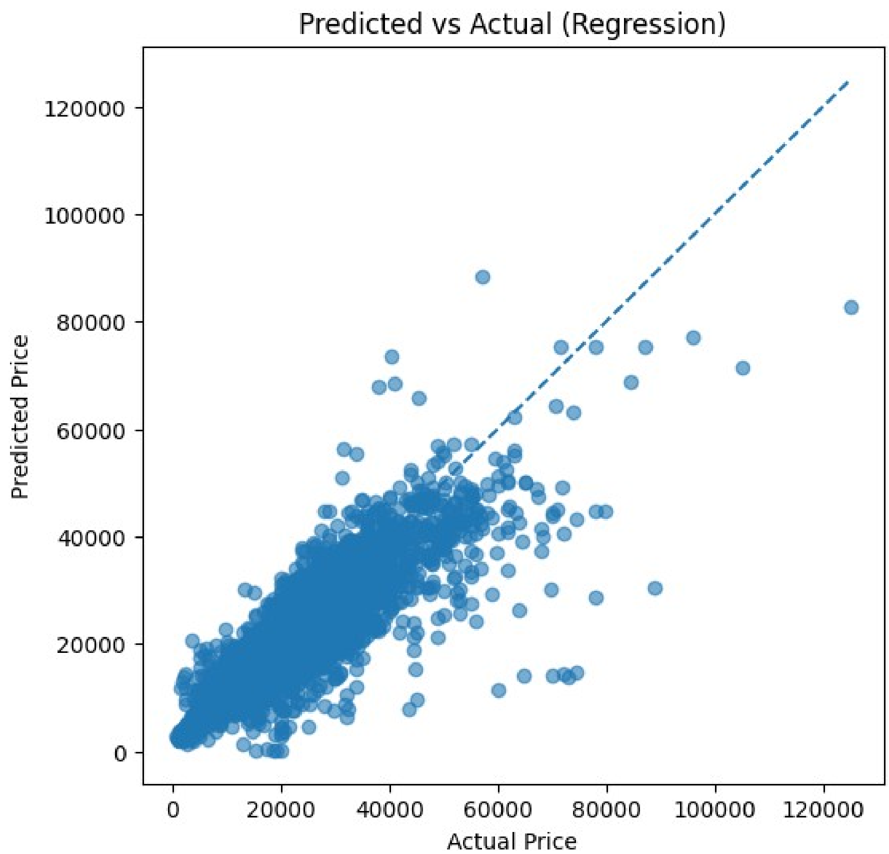

Figure 4. Predicted vs actual car prices for the regression model.

5. Assignment 2: Baseline Football Models

Assignment 2 builds baseline models for football market value prediction and performance-tier classification. The project compares polynomial regression, regularized regression, logistic regression, and Naive Bayes models.

Baseline Area	Best / Main Finding
Polynomial regression	Best polynomial degree = 4; test R2 = 0.8747
Regularized regression	Best model selected for CV: Lasso
Lasso test result	RMSE = 2.6354; R2 = 0.8880
Ridge comparison	Ridge alpha = 10 had test RMSE = 2.6500 and R2 = 0.8868
Logistic Regression	L1 penalty performed best: test accuracy = 0.8808 and weighted F1 = 0.8807
Naive Bayes	GaussianNB was the best NB variant, but lower than Logistic Regression

Cross-Validation Result	Mean	Std
Best regression baseline: Lasso RMSE	2.1654	0.4292
Logistic Regression L1 accuracy	0.8848	0.0034
GaussianNB accuracy	0.7169	0.0071

6. Assignment 3: Advanced Football System

Assignment 3 expands the football modeling system by testing advanced models and ensembles. For regression, it compares KNN Regressor, SVM Regression, Random Forest Regressor, and a Voting Regressor Ensemble. For classification, it compares KNN, SVM, Random Forest, and a Voting Classifier Ensemble.

Advanced Regression Model	Test RMSE	Test R2	Diagnosis
KNN Regressor	0.1779	0.9461	Stable but weaker than other advanced models
SVM Regression	0.1216	0.9748	Relatively stable generalization
Random Forest Regressor	0.1226	0.9744	Relatively stable generalization
Voting Regressor Ensemble	0.1193	0.9757	Selected final regression model

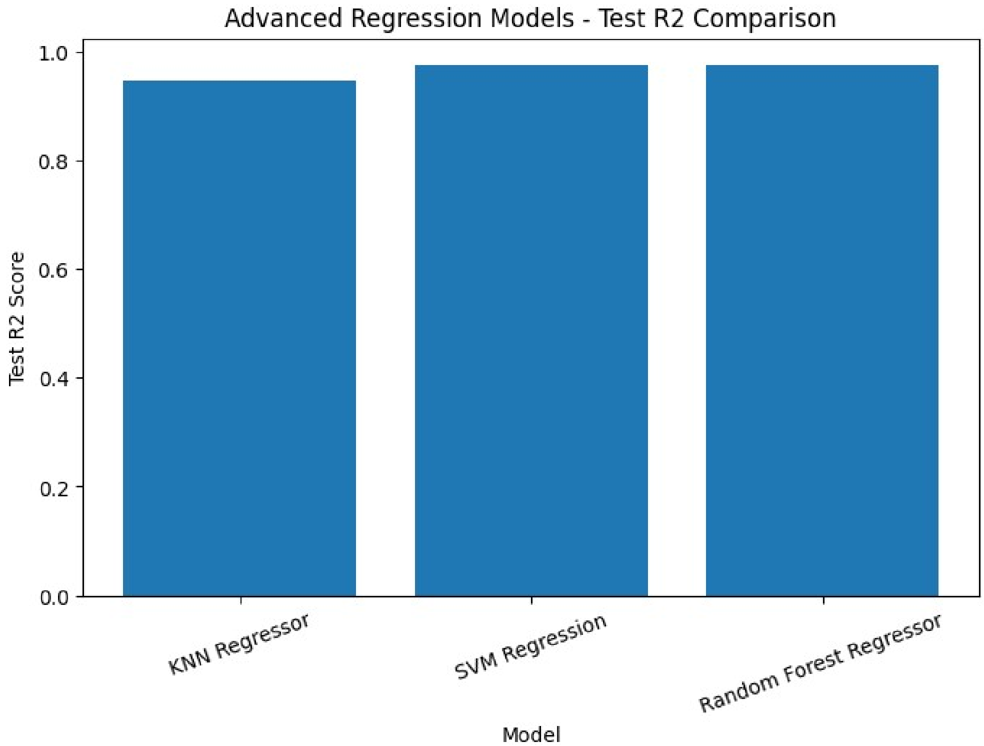

Figure 5. Advanced regression model comparison.

Advanced Classification Model	Test Accuracy	Test F1 Macro	Train-Test Gap
KNN Classifier	0.7659	0.7621	0.2340
SVM Classifier	0.8617	0.8602	0.0596
Random Forest Classifier	0.8457	0.8446	0.1498
Voting Classifier Ensemble	0.8493	0.8480	0.1469

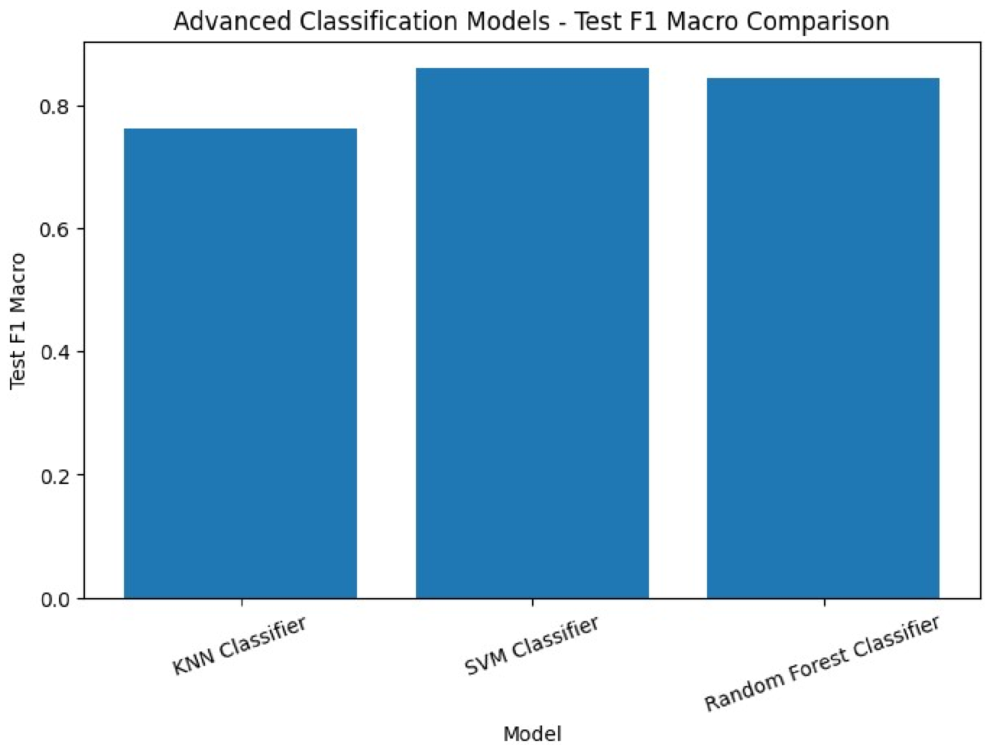

Figure 6. Advanced classification model comparison.

7. Final Model Selection and System Behavior

Final Component	Selected Model	Reason
Regression	Voting Regressor Ensemble	Best test R2 among the advanced regression models and strong CV stability
Classification	SVM Classifier	Highest test accuracy and the smallest train-test accuracy gap among the advanced classifiers

Final Stability Metric	Value
Voting Regressor Ensemble CV R2 mean	0.9681
Voting Regressor Ensemble CV R2 std	0.0078
SVM Classifier CV accuracy mean	0.8604
SVM Classifier CV accuracy std	0.0050

The unified inference function takes a new player profile and returns both a predicted market value and a predicted performance tier. Example project predictions were:

Country	Position	Age	Overall	Potential	Team	Total Stats	Predicted Value	Predicted Tier
Egypt	ST	24	82	86	Liverpool	1900	38.16 M$	Elite
France	CM	21	76	88	Paris Saint-Germain	1800	15.47 M$	Elite
Brazil	LW	29	89	89	Real Madrid	2100	65.33 M$	Elite

8. Assignment 2 vs Assignment 3 Comparison

Task	Assignment 2 Baseline	Assignment 3 Final	Change
Regression	Lasso CV RMSE = 2.1654 +/- 0.4292	Voting Regressor CV RMSE = 1.7900 +/- 0.4444	RMSE reduced by 17.34%
Classification	Logistic Regression L1 CV Accuracy = 0.8848 +/- 0.0034	SVM CV Accuracy = 0.8604 +/- 0.0050	Accuracy changed by -2.75%

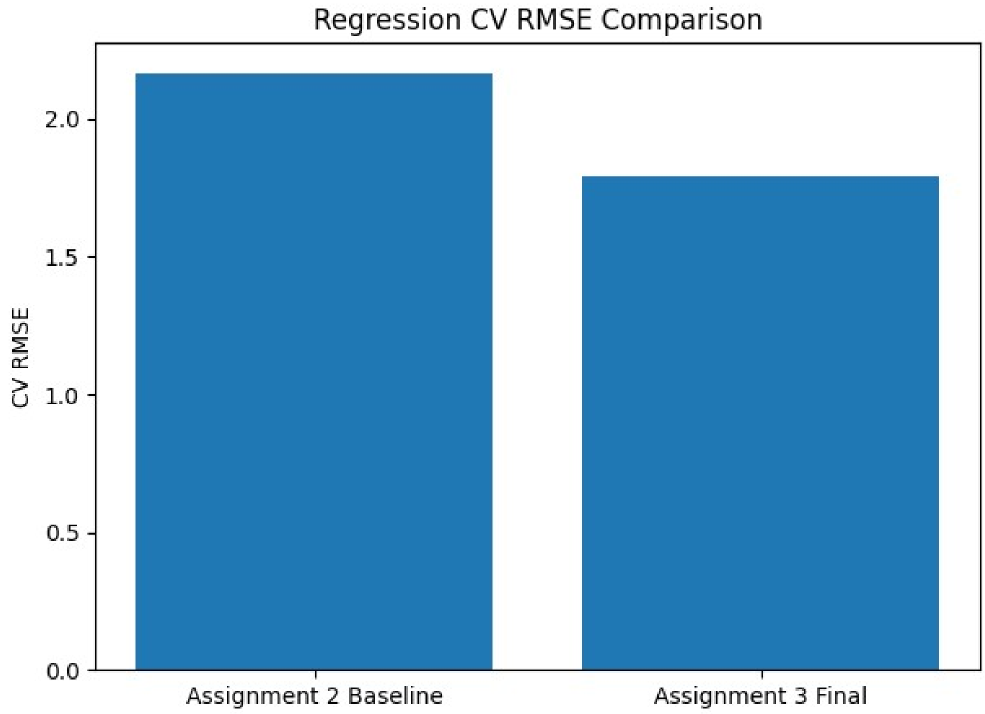

Figure 7. Regression RMSE comparison between Assignment 2 and Assignment 3.

9. Discussion

- The car price model produced a reasonable regression result, with an R2 near 0.80. The right-skewed price distribution and high variation in vehicle prices likely made the task harder.
- For the football project, market value was extremely right-skewed. This explains why preprocessing, regularization, and model comparison were important.
- Assignment 3 improved the regression system, especially by combining models through the Voting Regressor Ensemble. This reduced CV RMSE compared with the Assignment 2 Lasso baseline.
- Classification did not improve in Assignment 3 compared with Assignment 2. The Logistic Regression L1 baseline had higher CV accuracy than the final SVM classifier, although SVM still generalized well and had a small train-test gap.
- The final system is useful because it produces two outputs from one player profile: a predicted market value and a performance category.

10. Conclusion

Overall, the notebook demonstrates a full machine learning workflow across two applied prediction problems. It includes strong preprocessing, exploratory analysis, baseline modeling, advanced modeling, cross-validation, visual evaluation, and final inference testing. The football scouting system is the more complete final product because it combines regression and classification into a unified prediction system.

11. Recommended Improvements

- For the car price project, compare Linear Regression with stronger models such as Random Forest, Gradient Boosting, or XGBoost to improve RMSE.

- For the car price log-transform step, use np.log1p and np.expm1 together consistently to avoid small transformation mismatch issues.
- For the football project, report both scaled and original-unit regression metrics clearly, because Assignment 3 advanced regression results appear to use transformed/scaled target values.
- Keep the Logistic Regression L1 result in the final comparison, because it outperformed the SVM classifier in cross-validation accuracy.
- Add a short limitations section in the notebook explaining that market value can depend on external factors not in the dataset, such as contract length, injuries, league strength, and transfer demand.

Source: Generated from the uploaded Jupyter notebook file. Results and metrics were taken from the notebook outputs.